Huynh Quoc Viet

AI Research | *Machine Learning* | *Data Engineering*

Ho Chi Minh City, Vietnam | viethq1609@gmail.com | +84 932 764 532

LinkedIn | GitHub | Portfolio | ORCID

PROFESSIONAL SUMMARY

AI engineering student and research lead with work in conformal prediction, time-series forecasting, computer vision, and multimodal anomaly detection. Co-author of an accepted AJCAI 2026 poster. Built reproducible ML experiments and data pipelines using Python, PyTorch, scikit-learn, Pandas, SQL, FastAPI, and PostgreSQL.

EDUCATION

FPT University, B.Eng. in Artificial Intelligence — Jun 2023 – Expected Jun 2027

GPA: 3.12/4.0; Honorable Student of the Semester (Fall 2025).

HELP University, English Proficiency Program — Dec 2023 – Feb 2024

Kuala Lumpur, Malaysia.

TECHNICAL SKILLS

Programming & data: Python, SQL, Pandas, NumPy, PostgreSQL, Supabase.
Machine learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, Hugging Face, OpenCV, YOLO, RT-DETR.
Engineering: FastAPI, Streamlit, Docker, Linux, GitHub Actions, Weights & Biases, ONNX.
Research methods: Conformal prediction, time-series forecasting, multimodal anomaly detection, computer vision, reinforcement learning, reproducible evaluation.

EXPERIENCE

SpeedyLabX — Nov 2025 – Present

AI Tech Lead — AI Research — *Research Team*

- Co-authored research on air-quality alerts, multimodal anomaly detection, device-disjoint IoT classification, and sparse-monitoring bloom forecasting; contributed to experiment design and manuscripts.
- Designed leakage-resistant splits, calibration audits, and reproducibility checks for machine-learning evaluations.

Ky Nguyen So (General Era Digital Solution JSC) — Jan – May 2026

AI Research Intern — FPT University OJT — *Ho Chi Minh City*

- Reviewed federated, continual, and federated continual learning research; compared class-, domain-, and task-continual settings, data heterogeneity, forgetting, and edge constraints.

Outlier — Project Aether — Jun – Oct 2026

Freelance AI Evaluation Contributor via Upwork — *Remote*

- Labeled and validated Vietnamese AI evaluation data; compared spoken responses from two models using audio quality, naturalness, utility, and conversation-dynamics rubrics.

SELECTED RESEARCH

Warning Before the Evening Peak: Health-Centred Air Quality Alerts in New Zealand and Australia

AJCAI 2026 — Accepted Poster

- Developed exposure-weighted 1–3 day alert models using 10 years of hourly Canterbury observations and four years of Melbourne data; one-day MAE was 3.19 $\mu g/m^3$ and R^2 was 0.54 in the archived-forecast experiment.

Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution

KDD 2027 — Decision Pending

- Built the C-DLE framework and evaluated seven Time-MMD domains; in a controlled influenza study at conformal level 0.10, fusion increased F1 from 0.641 to 0.757 and TPR from 0.715 to 0.978.

SELECTED RESEARCH (CONTINUED)

Distribution-Free Coverage of Attack Classes on Devices Absent from Training
AISTATS 2027 — Under Review

- Tested five classifiers and four conformity scores on 68 physical IoT devices with device-disjoint splits; a device-quantile rule kept at least 92.1% of unseen devices within tolerance. A 5% target is unattainable with only 14 calibration devices.

Stress-Testing Multi-Source Cyanobacterial Bloom Forecasting under Sparse Monitoring
ACIIDS 2027 — Under Review

- Built a site-held-out benchmark across seven lakes; on the untouched 2023–2024 observed-target holdout ($n = 170$), seven-day chlorophyll-a history outperformed full fusion (weighted F1 0.791 vs. 0.688; adjusted $p = 0.016$).

SELECTED PROJECTS

BioListen VN — Vietnam AI Innovation Challenge Jul 2026
AI & Technical Lead, NeuraX.ai *Top 10 — Stage 1*

- Led a three-person team and built a 4,741-sample audio pipeline (3,792 train / 949 validation) for three ecological groups and nine threat/background classes.
- Trained a dual-head EfficientNet-V2-S with masked multi-task loss; added ONNX export and Grad-CAM inspection.

Personal Protective Equipment Detection Sep – Dec 2024
Deep Learning Engineer *Course Project*

- Harmonized nine public datasets into a nine-class detection schema; benchmarked YOLOv8s/m/l and RT-DETR models.
- Built a Streamlit interface for image, webcam, and video inference with compliance logic and annotated-video export.

FitSync — Personal Training Workspace Sep 2026 – Present
Project Lead — Product & OCR *Startup Course Project*

- Lead a five-member team building a Next.js and Supabase/PostgreSQL workspace with tenant-isolated profiles, row-level security, private meal-photo records, and progress history.
- Developing an OCR workflow with private evidence storage, required coach review, and manual fallback; benchmark design is in progress.

Adaptive Preference Allocation for Multi-Objective RL Sep 2026 – Present
Research Engineer *Course Project*

- Implemented a preference-conditioned vector Q-network with Double-DQN targets, adaptive allocation, replay provenance, and reproducible experiment orchestration; multi-seed experiments remain in progress.

NASA Space Apps 2025 — CosmicOptic 2025
Data & Model Engineer *Hackathon Project*

- Built a Lightkurve pipeline for Kepler/K2 light-curve acquisition, stitching, and normalization; engineered transit/orbital features and integrated FastAPI visualization.

Face Recognition Attendance System Jun – Aug 2024
Model Engineer *Course Project*

- Built an OpenCV/Tkinter face-data pipeline with 20 captures per person, image preprocessing, Haar Cascade detection, LBPH recognition, and CSV attendance records.

CERTIFICATIONS

IBM Data Science Fundamentals with Python and SQL; NVIDIA Building RAG Agents with LLMs.